

Assignment 1 Report

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Part 1: Coin Detection, Segmentation, and Counting

Objective

The goal was to detect, segment, and count the number of scattered Indian coins in an image using computer vision techniques.

Approach

The following steps were followed to achieve the objectives:

1. Edge Detection for Coin Detection

- **What was tried?**
 - Used Canny edge detection to detect coin boundaries after applying preprocessing techniques.
 - Experimented with different thresholds for Canny edge detection.
- **What worked?**
 - Using Gaussian blur before applying Canny edge detection helped reduce noise and improve detection.
 - The final thresholds (90, 255) provided the best results.
- **Final Approach:**
 - Convert the image to grayscale.
 - Apply Gaussian blur to smoothen the image.
 - Use Canny edge detection to identify the contours of the coins.
 - Draw contours around detected coins.

2. Coin Segmentation

- **What was tried?**
 - Applied region-based segmentation using contour detection.
 - Attempted morphological operations (erosion/dilation) to refine edges.

- **What worked?**
 - Contour detection successfully segmented individual coins.
 - Morphological operations were not required as Canny edge detection worked well.
- **Final Approach:**
 - Extract the contours detected from the edge-detected image.
 - Use `cv2.findContours()` to separate individual coins.

3. Counting Coins

- **What was tried?**
 - Used the length of detected contours to count the coins.
 - Experimented with filtering small contours to remove noise.
- **What worked?**
 - Directly counting the contours from edge-detected images gave accurate results.
- **Final Approach:**
 - The number of coins = total number of contours detected.

Final Implementation

- Reads an input image.
 - Converts it to grayscale.
 - Applies Gaussian blur and Canny edge detection.
 - Detects and outlines coins using contours.
 - Saves intermediate images for visualization.
 - Displays the final detected image along with the total coin count.
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Part 2: Image Stitching for Panorama Generation

Objective

To create a stitched panorama from multiple overlapping images using feature detection and homography transformation.

Approach

The following steps were followed:

1. Keypoint Extraction

- **What was tried?**
 - Used SIFT (Scale-Invariant Feature Transform) for keypoint detection.
 - Experimented with ORB and SURF but found SIFT to be more robust.
- **What worked?**
 - SIFT keypoints provided good matching results.
- **Final Approach:**
 - Convert input images to grayscale.
 - Detect keypoints and descriptors using SIFT.
 - Match keypoints between overlapping images.

2. Image Stitching

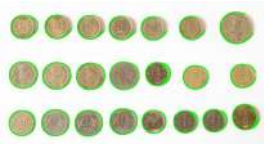
- **What was tried?**
 - Used Brute-Force Matcher (BFMatcher) with KNN matching.
 - Applied homography estimation using RANSAC to align images.
- **What worked?**
 - Using RANSAC filtering improved the accuracy of transformation.
 - Stitching in pairs first (left-center, then right) resulted in better alignment.
- **Final Approach:**
 - Match keypoints using BFMatcher and KNN matching.
 - Estimate homography using RANSAC.
 - Use cv2.warpPerspective() to transform images and blend them.

Final Implementation

- Loads three images.
 - Converts them to grayscale.
 - Extracts SIFT keypoints and descriptors.
 - Matches keypoints and finds homography transformation.
 - Stitches images in pairs and produces the final panorama.
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Observation Images

Task 1: Coin Detection, Segmentation, and Counting

Image Description	Observation Image
Original Input Image	
Grayscale Image	
Blurred Image	
Canny Edge Detection Result	
Final Coin Detection with Count	

Task 2: Image Stitching for Panorama

Image Description	Observation Image
Original Input Images (Left, Center, Right)	

Grayscale Images



Keypoints Detected in Each Image



Final Stitched Panorama



Conclusion

- **Part 1 (Coin Detection):** Successfully detected and counted Indian coins using edge detection and contour-based segmentation.
- **Part 2 (Image Stitching):** Successfully stitched images into a panorama using SIFT keypoints, RANSAC-based homography estimation, and image warping.